NTIRE 2026 Image Denoising ($\sigma = 50$) Challenge Factsheet DualExNet Hybrid Dual-Branch Image Denoising Network

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1. Factsheet Information

1.1. Team details

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- **Affiliation with NTIRE 2026 sponsors:** None
- **CodaBench user name and submission entry:**
  - User name: mayukun
  - Submission ID: 626240
- **Best scoring entry during development/validation phase:** PSNR 28.10 dB
- **Best scoring entry during test phase:** PSNR 29.61 dB
- **Link to codes:https://github.com/Entropyw/DualExNet.git**

1.2. Method details

As illustrated in Fig. 1, in this paper, a new Dual-Branch Exchange Network (DualExNet) is proposed. In consideration of the double dependency of the image denoising task on local high-frequency information and global low-frequency information, a parallel dual-branch architecture is employed in the network.

The forward propagation of the proposed model employs a parallel dual-stream architecture: the local feature branch extracts fine-grained local features via Residual Swin Transformer Blocks (RSTB)[3], while the global feature branch captures global context utilizing a U-Net structure[4]. Notably, the Bi-directional Multi-scale Feature Exchange Module (BIMFE) module is integrated at the skip connections between the encoder and decoder, facilitating the adaptive fusion and residual enhancement of cross-scale (Top-Down/Bottom-Up) features through learnable gating mechanisms. Subsequently, the multi-scale intermediate features from both branches are fed into the Hybrid Multi-scale Feature Exchange Module (HMFEM) module, where they undergo spatial alignment, channel concatenation, and progressive upsampling reconstruction to generate a global residual correction map. The final output is obtained by dynamically weighting the predictions of the dual branches using pixel-level weights generated by an adaptive gating network, followed by the addition of the HMFEM residual. Furthermore, a self-ensemble strategy encompassing 8 geometric transformations is introduced during the in-

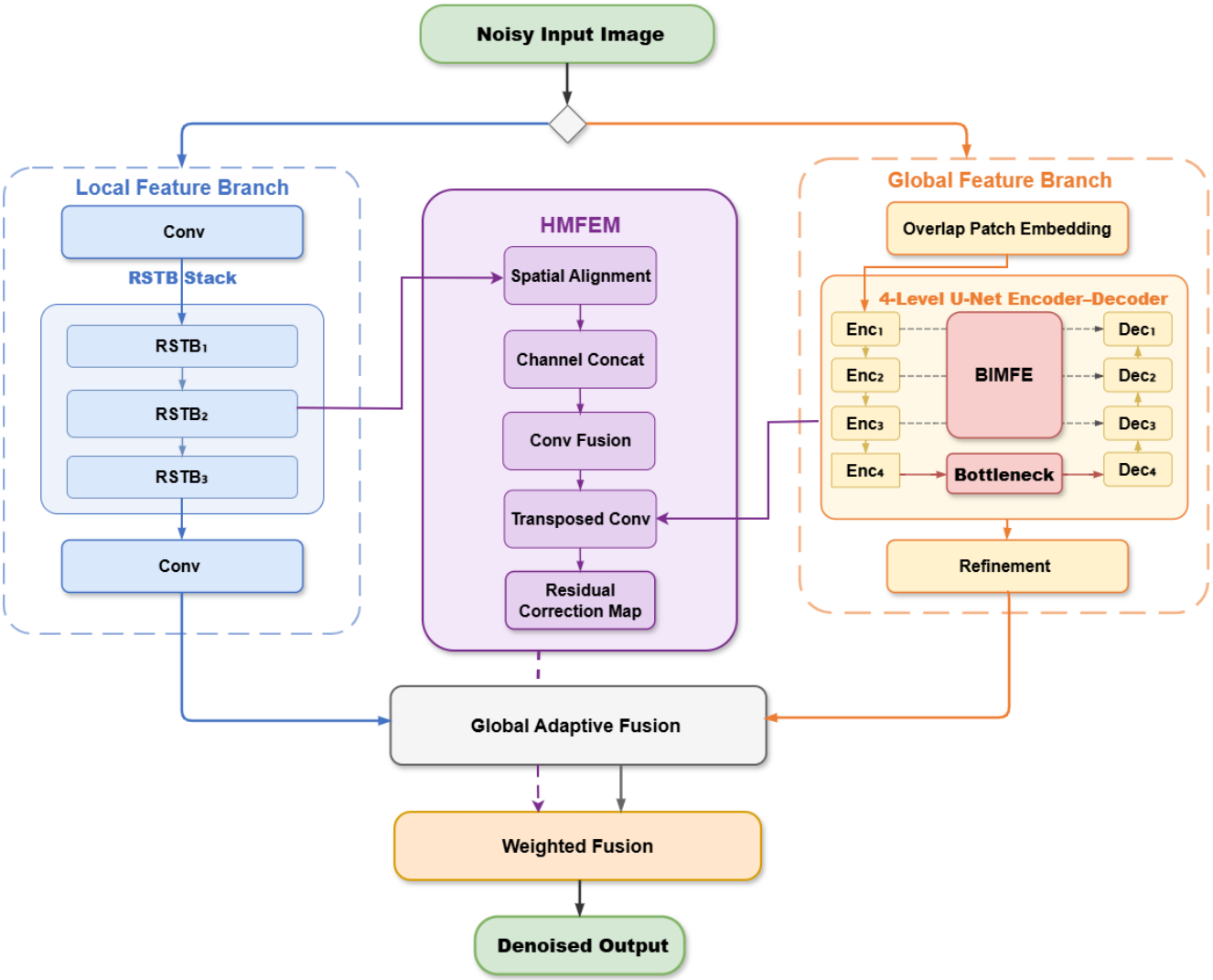

Figure 1. Overall architecture of the proposed DualExNet .

ference phase, which further boosts the denoising performance and robustness of the model through multi-view prediction fusion.

1.2.1. Bi-directional Multi-scale Feature Exchange Module

In traditional hierarchical feature reconstruction, single-scale feature fusion often leads to local loss of multi-scale contextual information. To introduce cross-scale information flow and enhance inter-layer feature interaction, we integrate BIMFE into the skip connections of the global feature branch.

For the BIMFE module at level i, its input is defined as the current level feature X_{main}, the high-resolution feature X_{prev} from the upper-level encoder, and the low-resolution feature X_{next} from the lower-level decoder. Its multi-scale fusion mechanism can be formally expressed as:

$$F_{fusion} = X_{main} + \alpha_{down} \cdot \mathcal{D}(X_{prev}) + \alpha_{up} \cdot \mathcal{U}(X_{next}) \tag{1}$$

where $\mathcal{D}(\cdot)$ denotes the downsampling operation based on 3×3 convolution (stride 2); $\mathcal{U}(\cdot)$ denotes the upsampling operation composed of 1×1 convolution and pixel shuffle. α_{down} and α_{up} are learnable gating weights initialized to 0.5, allowing the network to adaptively optimize the contribution of cross-scale feature flows during training.

To ensure the stability of gradient propagation in the network backbone (i.e., the integrity of the original feature flow) while incorporating multi-scale information, the fused feature F_{fusion} needs to sequentially pass through Layer Normalization (LayerNorm) and Attention mechanisms for

feature filtering. It is then superimposed on the current main branch feature X_{main} in the form of a residual:

$$X'_{out} = X_{main} + \text{Attention}(\text{LN}(F_{fusion})) \tag{2}$$

The final feature undergoes non-linear mapping through a Feed-Forward Network (FFN):

$$X_{out} = X'_{out} + \text{FFN}(\text{LN}(X'_{out})) \tag{3}$$

where X'_{out} represents the intermediate feature after attention modulation, and X_{out} is the final output of the BIMFE block.

1.2.2. Hybrid Multi-scale Feature Exchange Module

The core purpose of HMFEM is to eliminate the semantic differences between the local feature and global feature branches at the feature level, achieving deep fusion of local and global features. This module receives the hierarchical feature set $\{S_0, S_1, S_2\}$ from local feature branch and the corresponding feature set $\{R_0, R_1, R_2\}$ from global feature branch.

Due to differences in spatial resolution and channel dimensions, HMFEM first utilizes 1×1 convolution to perform channel projection on S_i, and performs spatial alignment via bilinear interpolation. Subsequently, concatenation is performed in the channel dimension, and deep aggregation is achieved through a 3×3 convolutional layer with GELU activation:

$$M_i = \text{GELU}\Big(\text{Conv}_{3\times3}\big([\text{Proj}(S_i), R_i]\big)\Big) \tag{4}$$

where M_i denotes the fused feature map at scale i, $\text{Proj}(\cdot)$ represents the channel projection and spatial alignment operation, and $[\cdot,\cdot]$ denotes the concatenation operation. The generated multi-scale feature representations $\{M_0, M_1, M_2\}$ will be progressively upsampled and fused through a bottom-up path based on transposed convolution . Finally, an output projection layer maps it to the residual correction map R_{hmfem} in the global image space.

1.2.3. Global Adaptive Fusion Module

To obtain the optimal output reconstruction result, this model abandons the naive average pooling strategy and designs an adaptive gating fusion mechanism based on feature content, which is employed in both training and inference phases. Specifically, the dynamic gating map is first generated by concatenating the two branch prediction maps P_{swin} and $P_{\text{restormer}}$ along the channel dimension. We then obtain the pixel-level dynamic gating weight map $G \in [0, 1]$ via a lightweight network with two convolutional layers and GELU, Sigmoid activations. Then, spatial weighted fusion is implemented by employing this weight map to conduct soft-gating combination on the two sets of predictions, formulated as:

$$P_{\text{gated}} = G \odot P_{\text{swin}} + (1 - G) \odot P_{\text{restormer}} \tag{5}$$

where $\odot$ denotes element-wise multiplication. Finally, we perform high-frequency residual compensation by accumulating the residual features from the High-Frequency Multi-Feature Enhancement Module (HMFEM) to the fused image, yielding the final denoised output with refined texture.

1.2.4. Train Strategy

We train our model on a combined dataset comprising 800 high-resolution images from DIV2K [1] and 84,991 images from LSDIR [2], with a total of 85,791 training pairs.This large-scale training data helps ensure robust generalization across diverse scenes. The training process is supervised using the Mean Absolute Error (MAE), i.e., $\mathcal{L}_1$ Loss, which effectively constrains pixel-level reconstruction fidelity. We employ the AdamW optimizer with default parameters $\beta_1 = 0.9$, $\beta_2 = 0.999$, and a weight decay of 1×10^{-4} to minimize the loss function. The initial learning rate is set to 1×10^{-4} and is scheduled using the Cosine Annealing strategy, gradually decaying to a minimum of 1×10^{-7} over the entire training cycle.During the inference phase, we employ a self-ensemble strategy to improve denoising performance. We perform 8 geometric transformations on the input image, consisting of horizontal/vertical flipping and rotation at $0°, 90°, 180°, 270°$. The final denoised result is obtained by averaging the inversely transformed predictions of all augmented images,.

1.2.5. Experimental Results

The objective quality evaluation of the model employs Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity (SSIM) as core metrics. Based on the validation set and test set provided officially by the NTIRE 2026 Challenge, the quantitative performance of this model is shown in Tab. 1:

Table 1. Quantitative performance on NTIRE 2026 dataset.

Dataset (NTIRE 2026)	PSNR (dB)	SSIM
Validation Set	28.10	0.79
Test Set	29.61	0.86

2. Other details

- Planned submission of a solution(s) description paper at NTIRE 2026 workshop.
- Owing to time limitations in the validation stage, the results were derived from a model that had not been fully trained.

References

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